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A MINDFUL ALTERNATIVE TO SCREEN TIME: THE SHORT-TERM EFFECTS OF A BREATH-FOCUSED MINDFULNESS EXERCISE VS. UNSTRUCTURED SMARTPHONE SCREEN TIME ON HEART RATE VARIABILITY AND RELAXATION

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Introduction: Many smartphone users spend hours daily on unstructured screen time leisure activities including social media, games, and videos. Given that many users express a desire to cut back on this habit, it is important to test the relative physiological and emotional benefits of alternative activities, such as mindfulness meditation, as daily practices for relaxation and restoration. **Methods**: This study tests the immediate effects of unstructured mobile phone screen time vs. practice of a brief mindful breathing meditation exercise on heart rate variability (HRV, a common measure of healthy resting physiological activity) and self-reported states of relaxation. Participants (171 female university students) were randomly assigned to 15 minutes of unstructured screen time (i.e., use phones in a manner consistent with typical daily use) or a mindfulness induction (i.e., a guided meditation recording focused on mindfulness of breathing). **Results:** Consistent with hypotheses, HRV showed greater increases relative to baseline in the mindfulness condition than the screen time condition (High-frequency HRV, partial $\eta^2 = .14$, $p < .001$; Root Mean Square of Successive Differences, partial $\eta^2 = .053$, $p = .002$). Subjective relaxation increased in both conditions (partial $\eta^2 = .47$, $p < .001$); however, mindfulness produced greater increases in relaxation over time than unstructured screen time (partial $\eta^2 = .089$, $p < .001$). **Conclusions:** Mindfulness was superior to unstructured screen time at increasing physiological and subjective markers of relaxation. Results suggest that repurposing smartphones to facilitate mindfulness practice though guided meditation apps may be a more effective method of promoting physiological and emotional well-being than more ubiquitous unstructured smartphone use.

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According to consumer surveys, adults in the United States are estimated to spend an average of 4.23 hours per day using mobile devices such as smartphones (Wurmser, 2021), which reflects a roughly 14-fold increase over the prior 12 years (cf. Meeker, 2018). Mobile phones are seamlessly integrated into daily life and can be used for a variety of recreational, work, and educational activities. Unstructured use of mobile phones for leisure purposes such as checking social media, playing video games, and watching videos (sometimes referred to as *screen time*) is a common way that people pass short periods of unscheduled time in daily life. Although screen time activities such as social media may be motivated in part by a desire to relax and reduce negative emotions (Ni et al., 2020), it is unclear if screen time is in fact relaxing or emotionally restorative. Concerns have been raised about the cumulative adverse mental health effects of excessive screen time, especially for young people. Large population-based studies of adolescents and young adults report an association between excessive mobile phone use and emotional distress (Twenge et al., 2018), which may be due in part to the displacement of non-digital activities such as physical activity, sleep, and in-person social interactions. A Pew survey found that 54% of teens believe they spend too much time on their phones (Jiang, 2018). In a separate survey of mobile phone users, nearly two-thirds indicated that they would like to reduce their phone use; however, only half of these users reported that they were successful in doing so (Deloitte, 2018).

There is growing evidence that cutting back on common mobile phone activities such as social media can have mental health benefits. Social media users randomly assigned to either reduce their social media use completely (Tromholt, 2016) or cut back to less than 30 minutes per day (Hunt et al., 2018) showed greater improvements in happiness and reductions in depression and loneliness one to two weeks later compared to those who maintained consistent use. This evidence for the benefits of behavior change around technology habits can be persuasive for individuals looking to reduce their screen time; however, such results cannot speak to potential immediate benefits of choosing an alternative activity over mobile phone use, information that could further help to motivate behavioral change.

MINDFULNESS IN THE CONTEXT OF PHONE USE

Mindfulness practice has been described as a promising alternative to excessive or problematic mobile phone use. Mindfulness reflects nonjudgmental awareness of present-moment experiences and has been operationalized as a temporary state as well as both a dispositional tendency and an enduring effect cultivated through meditation and other forms of mindfulness practice (Keng et al., 2011). When mobile phones first started to become a ubiquitous part of daily life, pioneering mindfulness teacher and researcher Jon Kabat-Zinn (2005) suggested that promoting mindfulness would help to lower individuals' harmful over-reliance on mobile communication technology by increasing awareness of immediate internal and external experiences and reducing the need to use technology to avoid or suppress internal experiences. Subsequent research has demonstrated that individuals who score low in dispositional measures of mindfulness are more likely to demonstrate problematic mobile phone habits including texting-while-driving and using their phones to regulate their emotions (Feldman et al., 2011). In addition, excessive mobile phone use is associated with greater sleep disturbance among individuals scoring low in dispositional mindfulness (Liu et al., 2017), and lower dispositional mindfulness mediates the relationship between social media engagement and depression symptoms (Jones et al., 2022). Brewer (2019) characterized various forms of technology use including smartphones, social media, and video games as potentially addictive experiences and presented a neurobiological rationale for how mindfulness practices may be uniquely suited to assist with modifying these behavioral patterns.

In terms of putting these ideas into practice, an individual could temporarily re-purpose their phone to support practicing mindfulness by following a guided meditation recording through a smartphone-based app as a competing behavior to unstructured mobile phone time. Randomized controlled trials have established a substantial body of evidence that mindfulness-based interventions produce reductions in emotional distress in a wide variety of non-clinical samples (Khoury et al., 2015) including university students specifically (Dawson et al.,

2020) and can be effectively delivered via mobile phone apps (Gál et al., 2021). Furthermore, even brief periods of mindfulness meditation have been shown to produce immediate reductions in emotional distress in laboratory experiments (Leylard et al., 2018). Experimental studies examining the effects of single sessions of specific mindfulness practices add valuable perspectives to the literature by complementing findings of multicomponent interventions presented in randomized controlled trials (Feldman et al., 2010, Keng et al., 2011). Such findings may be particularly relevant in informing research on the experiences of novice meditators who are informally exploring brief mindfulness practices such as those presented through apps.

PHYSIOLOGICAL MEASURES OF MINDFULNESS

In addition to findings on subjective measures of emotional state, several lines of evidence suggest that mindfulness practice is associated with increased heart rate variability (HRV), a physiological marker that has become a growing focus in research on mindfulness-based interventions (Christodoulou et al., 2020). HRV is the beat-to-beat variation in heart rate that is influenced by a complex interplay of various inputs including sympathetic and parasympathetic branches of the autonomic nervous system as well as respiratory sinus arrhythmia, the acceleration and deceleration of the heart caused by inhalation and exhalation, respectively (McCarty & Shaffer, 2015). Increases in HRV during mind-body practices such as meditation have been conceptualized as an objective measure of the relaxation response (Nakao, 2019). More broadly, HRV has been conceptualized as a marker of an individual's capacity to modulate cardiac activity in order to rapidly and flexibly respond with appropriate levels of arousal to evolving social and environmental demands (Laborde et al. 2017; McCarty & Shaffer, 2015). As such, HRV can be understood to reflect the psychological capacities that may be enhanced by mindfulness practice including self-regulation of attention and emotion (Christodoulou et al., 2020). The importance of identifying practices that can increase HRV is further underscored by research linking low levels of HRV to various forms of psychopathology (Beauchaine & Thayer, 2015) and deficits in emotion

regulation (Appelhans & Luecken, 2006; Williams et al. 2015) and self-control (Zahn et al., 2016).

In addition to the relevance of HRV to psychological and physical well-being, there is evidence that HRV metrics recorded during meditation practice are both a promising marker of mindfulness processes and a clinically useful predictor. HRV scores during meditation are associated with a behavioral measure of attention to breath sensations, a marker of successful engagement with mindfulness meditation (Burg et al., 2012). In terms of clinical utility of change in HRV, in a study of individuals participating in a mindfulness-based smoking cessation intervention, those who experienced the greatest improvement in HRV while meditating during a post-intervention laboratory assessment reported smoking fewer cigarettes at a follow-up assessment approximately 3 months later (Libby et al., 2012).

HRV IN THE CONTEXT OF MINDFULNESS PRACTICE AND PHONE USE

Several studies have tested the effect on HRV produced by multi-session mindfulness meditation training in young adults. For instance, undergraduate students participating in a 4-week mindfulness meditation stress-management group had significantly higher HRV during a cognitive challenge task administered after the treatment compared to participants in two control conditions: an ecologically valid group study break and no intervention (Shearer et al., 2016). Similarly, university students in a self-guided mindfulness training program evidenced greater increases in HRV at a follow-up assessment than those assigned to either a progressive muscle relaxation training or no-training group (Ditto et al., 2006). However, mixed results have also been observed in terms of the effects of mindfulness-based interventions on HRV (for reviews, see Brown et al., 2021; Christodoulou et al., 2020), suggesting this is a topic of investigation warranting further research.

Beyond the effects of multi-session mindfulness interventions, laboratory experiments have also shown that single sessions of mindfulness meditation practice among young adult novice meditators cause greater increases in HRV than

a variety of control conditions including listening to an audio recording describing mindfulness without practice instructions (Azam et al., 2015), practicing a progressive muscle relaxation exercise (Ditto et al., 2006, Study 1), listening to excerpts from popular audiobooks (Ditto et al., 2006, Study 2), and a sham relaxation procedure (May et al., 2016). However, comparisons of mindfulness practice to unstructured screen time have not been reported in the literature to date. Given the ubiquity of recreational mobile phone use in daily life, such a comparison would provide a particularly ecologically valid test of the effect of mindfulness practice on HRV as well as helpful information for individuals seeking alternative activities to recreational mobile phone use.

The effects of smartphone use on HRV is a less studied area. There is some correlational evidence that lower resting HRV is associated with higher scores on measures of internet addiction in children (Lin et al., 2014) and adults (Moretta et al., 2019) as well as internet gaming addiction in adolescents (Kim et al., 2016). These results suggest excessive engagement with internet-based activities is associated with poorer HRV; however, experimental studies are needed to support any causal effects of user-selected unstructured screen time activities. There is some evidence that typical mobile phone activities can increase physiological arousal. Sending text messages has been shown to increase heart rate, respiration, and skin conductance relative to resting baseline assessments (Lin & Peper, 2009). Naturalistic use of a mobile phone in the face of social stress can produce a relatively greater increase in sympathetic activity (measured with salivary alpha amylase) than having one's phone present but not in use (Hunter et al., 2018). However, we are not aware of any laboratory studies examining the immediate effects of naturalistic mobile phone use on HRV specifically.

CURRENT STUDY

In this study, we tested the immediate effects of unstructured mobile phone screen time vs. practice of a brief mindful breathing meditation exercise on HRV. Given the conceptualization of increased HRV as a relaxation response, we supplemented the assessment of HRV with a self-report assessment of changes in

states of relaxation in response to both meditation and mobile phone use. University students were randomly assigned to one of two conditions. In the phone condition, participants were prompted to use their mobile phones in a manner consistent with their typical daily use to visit websites and social media, use apps, play video games, send emails or texts, watch videos, or listen to music. In the meditation condition, participants were prompted to follow a recorded guided meditation focused on mindfulness of the breath. HRV was measured during a baseline assessment and then for the duration of the 15 minutes of either phone use or meditation. Self-reported relaxation was measured at the start of the study and immediately after the 15-minute activity. Based on theory and prior research, we hypothesized that brief mindfulness meditation practice would increase HRV and subjective relaxation more than unstructured mobile phone screen time. As an exploratory analysis, we also assessed the effect of the two activities on subjective anxiety.[1]

METHODS

PARTICIPANTS

The sample consisted of undergraduate students who were recruited from psychology classes at a women's university in the United States. One-hundred and eighty-two participants consented to participate in the study. Data from seven participants were excluded due to errors in collecting electrocardiogram (ECG) data or administering the experiment procedures. Due to documented gender differences in HRV (Koeing & Thayer, 2016), data from four participants (1 who identified as male, 3 who did not answer the gender question) were removed from analyses. The final sample consisted of 171 female participants (Age: M = 19.41, SD = 2.41; Race/Ethnicity: 74.9% White, 17.5% Asian or Pacific Islander, 2.3% Black or African American, 2.3% Other, 2.9% Selected two or more; 95.3% Non-Hispanic, 4.1% Hispanic, .6% Did not answer). Study participants received course credit in exchange for their participation.

1. We thank the anonymous reviewer who recommended this analysis.

PROCEDURES

Overview. Data collection took place in individual sessions. Participants were randomly assigned through simple randomization in their order of participation to one of two experimental task conditions: mindful breathing meditation exercise or unstructured mobile phone screen time (for the sake of brevity throughout the manuscript, these conditions are referred to as *meditation* and *phone*, respectively). At the start of the study, all participants were instructed to turn their mobile phone to silent mode and place it in their bag or pocket.

Participants were seated in a comfortable chair in front of a desk with a desktop computer and monitor. Participants first completed a pre-task mood assessment presented via a Qualtrics survey. Participants were then fitted with three disposable electrodes in a Lead II configuration with the positive electrode placed on lower left leg, negative electrode on right forearm, and grounding electrode on the right lower leg (Biopac Systems Inc., n.d.). Once electrodes were affixed, participants were instructed to relax and breathe normally, to remain seated during the recording with their palms facing up and feet on the floor, and to avoid movement or talking during the recording period. ECG was recorded during a baseline period and the duration of the experimental task (meditation or phone). After completing the task, participants completed a post-task mood assessment and demographic questions in a Qualtrics survey.

The study was approved by the Simmons University Institutional Review Board before data collection commenced. Participants completed written informed consent procedures before participating.

Baseline assessment. We utilized a "vanilla" baseline assessment in which participants' attention is minimally engaged via a standardized, cognitively non-demanding task to minimize introducing confounds of individual differences in pre-task anticipatory anxiety, intrusive thoughts, or boredom (Jenning et al., 1992; Piferi et al., 2000). Jennings and colleagues (1992) demonstrated that "vanilla" baseline assessments produce more stable baseline cardiovascular assessments than standard resting baselines. For the present study, we created a 7-minute video slideshow consisting of images of inanimate objects and scenes

drawn from the Geneva Affective PicturE Database ([GAPED]; Dan-Glauser & Scherer, 2011) that had been classified as neutral images in that participants in a stimuli norming study rated the images as low in arousal and neutral in valence (i.e., toward the middle of a negative-to-positive valence scale). The images were made into a slideshow video file in which each image was displayed for 8 seconds. The video was embedded in the Qualtrics survey. Participants were asked to watch this short video while the initial ECG recording was being made. The use of a "vanilla" baseline task with visual stimuli is consistent with prior research (Jennings et al., 1992; Piferi et al., 2000), and the presentation of images on a computer monitor adds ecological validity given that college students frequently use computer screens to view assigned course material and complete assignments. As such, the transition from viewing images on a computer screen to using a mobile phone or meditating would be analogous to taking a study break from schoolwork. An ECG recording was made for the full 7-minute duration; however, only the final 5 minutes of data were analyzed to allow time for participants to habituate to the recording environment (Laborde et al., 2017).

Task instructions

Unstructured Phone Screen Time Condition (Phone). Participants were asked to remove their mobile phone from their pocket or bag and hold it in their right hand. Participants were instructed to "use your phone in whatever way you would normally use it in your daily life whenever you have a bit of free time" with a few exceptions. Specifically, participants were invited to visit websites and social media, use apps, play video games, send emails or texts, watch videos, or listen to music. We provided participants considerable latitude in choice of activities to increase the ecological validity of capturing typical phone-based leisure activity of college students as a kind of "activity as usual" condition that would be conceptually analogous to a treatment as usual condition in a clinical study. To avoid potential confounds, participants were asked to refrain from making voice calls or using the phone's camera function including video messages. Participants were offered the use of headphones or a charger if they needed to borrow one. After these instructions, participants were told that they had 15 minutes to use their phone; the ECG recording would continue during this time, and the experimenter would

let them know when it is time to move onto the next activity. A summary of these instructions was presented on the video monitor during the 15 minutes so the participant could refer to them as needed.

Mindful Breathing Meditation Exercise (Meditation). Participants were asked to listen to a 15-minute recording that would guide them through a meditation exercise. The recording was embedded in the Qualtrics survey. Participants were invited to use either their own earbuds or headphones provided in the laboratory. After a sound check to confirm the participant could hear the recording and the volume was comfortable, the recording was started. The guided mindful breathing meditation recording is the same recording and procedure used in a prior laboratory experiment and found to differentially produce greater levels of decentering, a conceptually relevant outcome of mindfulness practice, when compared to other stress-management activities (see Feldman et al., 2010 for details). Briefly, the recording was narrated by a licensed clinical psychologist with extensive experience in teaching mindfulness meditation. It consisted of 12 minutes of guided instruction in the technique followed by a 3-minute period of silence before which participants were instructed to continue with self-guided practice. The recording began with instructions for participants to settle into a comfortable sitting position, sit with their back straight against the back of the chair with legs uncrossed, feet flat on the floor, and hands in their lap. In the recording, participants were guided to become aware of physical sensations—especially those associated with the process of breathing—and to observe them without the intention of altering them. Participants were asked to notice in an accepting, non-judgmental manner when their minds wandered to something other than the exercise and to gently return focus to the sensations of breathing when this occurred. During the 3 minutes of silence, participants were encouraged to continue to observe their breathing consistent with the guidance presented in the narrated portion of the exercise. Participants were told the activity would last 15 minutes, the ECG recording would continue during this time, and the experimenter would let them know when it was time to move on to the next activity.

MEASURES

Heart Rate Variability. HRV was measured by recording electrocardiographic activity (ECG) using a Biopac MP36 system. Signals were sampled at 1000 Hz and submitted to a 0.05-Hz highpass filter before being analyzed. ECG data were analyzed using Kubios 2.1 software. This software imports raw ECG data and visually presents each heartbeat captured by the original ECG recording as well as each R-R interval, the time in milliseconds elapsed between two successive heartbeats. After signal pre-processing, each file was reviewed for artifacts. First, the Kubios algorithm to detect artifacts was applied to each file with the algorithm sensitivity set to medium, which flags instances of a 25% or greater BPM change between beats. These artifacts are flagged on the R-R interval graph to allow visual inspection of the corresponding section of the ECG recording. Second, because this algorithm can sometimes flag actual heartbeats as artifacts (Laborde et al. 2017), a research assistant visually inspected each detected artifact to distinguish actual heartbeats to be retained vs. artifacts to correct (e.g., heartbeats that were detected inaccurately during pre-processing or technical artifacts such as temporarily lost signal due to electrode slippage or excessive participant motion). Logs made by experimenters of recording abnormalities in each data collection session were consulted to help with such determinations.

Data from most participants contained no artifacts requiring correction (n = 164, 94.8%). In the remaining cases (n = 7), fewer than 1% of beats (range .05 – .53%) required correction. Artifacts were corrected both manually (e.g., marking a beat that was not initially detected) and by using the Kubios artifact correction algorithm (see Tarvainen et al., 2021). Once artifacts were reviewed and corrected in indicated cases, HRV values were calculated by the Kubios software. The primary HRV metric selected was High Frequency HRV (HF-HRV) absolute power (0.15–0.40 Hz) computed using the Fast Fourier Transform method. We selected this metric in light of the common use of this and other frequency-based HRV assessments in prior studies examining acute effects of meditation on HRV (Azam et al. 2015; Libby et al., 2012; May et al., 2016). To complement this frequency-based measure of HRV, root mean square of successive differences (RMSSD) was

also analyzed as a time-domain measure. Laborde and colleagues (2017) encourage the reporting of both frequency and time-domain based measures of HRV and recommends HF-HRV and RMSSD specifically among the many HRV metrics available as markers of vagal tone or parasympathetic dominance. For each participant, four HRV values were extracted: baseline (final 300 seconds of 7-minute baseline recording) and three time periods during the 15-minute task (Time 1: 0 – 300 sec, Time 2: 301 – 600 sec, Time 3: 601 – 900 sec) for both HF-HRV and RMSSD. HRV values were log10 transformed to normalize positive skew. HRV values were imported into SPSS 27 for statistical analyses.

Subjective Mood Assessment. Mood state was assessed after participants completed informed consent and again immediately post-task using a series of mood adjectives rated on a 7-point scale, ranging from "Do not feel this way at all" to "Feel this way very strongly," with a midpoint labeled "Feel this way moderately." This assessment procedure is modeled on one developed by Nezlek and colleagues (2008) and is informed by the circumplex model of affect (Posner et al., 2005). A subjective relaxation score was calculated by averaging responses to four items reflecting the "negative deactivation" quadrant of the affect circumplex (calm, relaxed, peaceful, content). Internal consistency of the measure of state relaxation was acceptable at both pre-task ($\alpha = .84$) and post-task ($\alpha = .86$) assessments. State relaxation was selected as the dependent variable because it is the subjective mood state most analogous to the parasympathetic activity reflected by the selected measures of HRV. As an exploratory analysis, we also examined the effect of the two activities on state anxiety by examining items reflecting the negative activation quadrant of the affect circumplex. These four items (nervous, tense, stressed, upset) demonstrated acceptable internal consistency at both pre-task ($\alpha = .76$) and post-task ($\alpha = .81$) assessments.

ANALYSIS PLAN

Baseline differences in HRV and mood state across the two experimental conditions were assessed using independent samples *t*-tests. We performed two-way repeated-measures ANOVAs to examine differences in HRV and subjective mood states between

the two experimental conditions (meditation vs. phone, representing the between-subjects factor) across repeated assessments (representing the within-subjects factor of time). Mauchly's Test of Sphericity was used to test whether the assumption of sphericity had been violated for each ANOVA analysis involving more than two repeated assessments. Partial η^2 was examined to measure effect sizes in ANOVA analyses and were interpreted using Cohen's (1988) suggested benchmarks (small = 0.01, medium = 0.06, large = 0.14).

RESULTS

DESCRIPTIVE STATISTICS

Descriptive statistics for all variables are presented in Table 1. Baseline HRV and mood scores did not significantly differ across the two conditions (all *p*s > .12).

HRV ANALYSES

We analyzed the HRV data using a mean difference in change approach. Bodner & Bliese (2018) recommend this approach for studies in which there are repeated post-intervention assessments, the temporal separation of assessments is short, and the dependent variable has high within-condition cross-time correlation (ϱ_{12}) that is close to the ratio of pooled within-condition standard deviations for the dependent variable at baseline and follow-up assessments (*k*). Each of these criteria were met in the present study for both conditions across all three post-intervention assessments (range of ϱ_{12} = .86–.90 for HF-HRV, .90–.93 for RMSSD; range of *k* values = .96–1.01 for HF-HRV, .91–.99 for RMSSD). As such, we calculated change scores (Δ) for all three post-assessment timepoints (baseline HRV values subtracted from post-task HRV values at T1, T2, and T3, respectively) and analyzed these ΔHRV scores as dependent variables.

First, we performed a two-way repeated measures ANOVA to examine differences in change in HRV between the two experimental conditions (meditation vs. phone, representing the between-subjects factor) across the three ΔHF-HRV values

TABLE 1. Descriptive Statistics for HRV and Mood Variables

	Phone (n = 86)		Meditation (n = 85)	
	M	*SD*	*M*	*SD*
HF-HRV baseline	2.90	0.49	2.78	0.54
HF-HRV T1	2.86	0.49	2.90	0.56
HF-HRV T2	2.83	0.50	2.88	0.57
HF-HRV T3	2.80	0.48	2.86	0.55
ΔHF-HRV T1	–0.04	0.24	0.12	0.25
ΔHF-HRV T2	–0.07	0.26	0.10	0.28
ΔHF-HRV T3	–0.10	0.23	0.09	0.28
RMSSD baseline	1.61	0.25	1.55	0.27
RMSSD T1	1.62	0.25	1.61	0.27
RMSSD T2	1.62	0.25	1.60	0.29
RMSSD T3	1.61	0.25	1.60	0.28
ΔRMSSD T1	0.01	0.10	0.05	0.11
ΔRMSSD T2	0.00	0.11	0.04	0.13
ΔRMSSD T3	–0.01	0.11	0.05	0.11
Relaxation (Pre-task)	4.02	1.35	4.14	1.24
Relaxation (Post-task)	4.65	1.20	5.40	1.09
Anxiety (Pre-task)	3.13	1.31	3.19	1.15
Anxiety (Post-task)	2.29	1.08	2.16	0.99

Note. HF-HRV = High Frequency Heart Rate Variability (ms^2); RMSSD = Root Mean Square of Successive Differences (ms). Values for all HF-HRV and RMSSD variables are log10 transformed. ΔHF-HRV and ΔRMSSD scores are calculated by subtracting baseline HRV scores from HRV scores at each time point. Baseline HRV and mood scores did not significantly differ across the two conditions (range of p values = .12 – .73).

during the task (representing the within-subjects factor of time). Mauchly's Test of Sphericity indicated that the assumption of sphericity had not been violated, χ^2 (2) = .73, p = .70. Results indicated a significant main effect of time, $F(2, 338) = 3.94$, $p = .02$, partial $\eta^2 = .023$, and condition $F(1, 169) = 27.41$, $p < .001$, partial $\eta^2 = .14$, but not Condition × Time interaction, $F(2, 338) = .30$, $p = .75$. partial $\eta^2 = .002$ (see Table 2). These results indicate that participants assigned to the meditation condition exhibited relatively greater increases in HF-HRV during the three periods of the experimental period than the participants assigned to the phone condition (see Figure 1A). The effect of condition was a

TABLE 2. Two-Way Repeated Measures ANOVA Testing the Effects of Meditation and Phone Use Conditions on Change in Heart Rate Variability (HRV) and Subjective Mood State

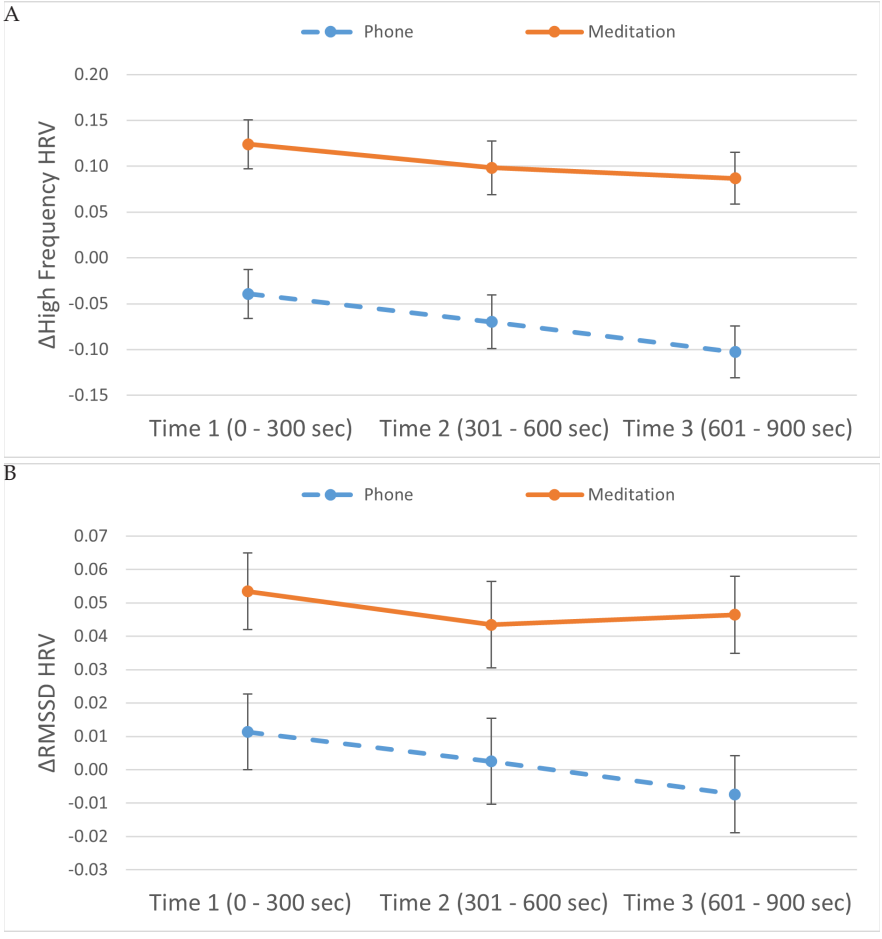
	df	*F*	Partial η^2	*P*
ΔHF-HRV				
Condition	1, 169	27.41	.14	< .001***
Time	2, 338	3.94	.023	.02*
Condition × Time	2, 338	.30	.002	.75
ΔRMSSD				
Condition	1, 169	9.54	.053	.002**
Time	1.95, 328.91	1.73	.010	.18
Condition × Time	1.95, 328.91	.490	.003	.61
Relaxation				
Condition	1, 169	6.70	.038	.01*
Time	1, 169	149.49	.47	< .001***
Condition × Time	1, 169	16.58	.089	< .001***
Anxiety				
Condition	1, 169	.04	< .001	.84
Time	1, 169	177.93	.51	< .001***
Condition × Time	1, 169	1.99	.012	.16

Note. ΔHF-HRV = Change in High-Frequency Heart Rate Variability, ΔRMSSD = Change in Root Mean Square of Successive Differences.

*$p < .05$. **$p < .01$. ***$p < .001$.

medium-to-large effect. The significant effect of time indicates that irrespective of group assignment, the increases in HF-HRV from pre-task baseline were greatest during the initial five-minute period (T1) but decreased in the subsequent periods (T2 and T3) and this effect of time represents a small effect.

To test the replicability of the ΔHF-HRV findings across a second metric of HRV, we performed a similar repeated measures ANOVA with ΔRMSSD as the dependent variable. Mauchly's Test of Sphericity indicated that the assumption of sphericity had been violated, χ^2 (2) = 7.78, p = .02. The Huynh-Felt corrections was applied because the observed epsilon value (.96) was greater than .75. Results indicated a significant main effect of condition, F(1, 169) = 9.54, p = .002, partial η^2 = .053, but not time, F(1.95, 328.91) = 1.73, p = .18, partial η^2 = .010, or Condition × Time



Note. ΔHF-HRV = Change in High-Frequency Heart Rate Variability; ΔRMSSD = Change in Root Mean Square of Successive Differences. HF-HRV and RMSSD values were log10 transformed and ΔHRV scores calculated by subtracting baseline HRV scores from HRV scores at each time point. Graphed values reflect estimated marginal means and bars represent one standard error.

FIGURE 1. Change in high-frequency HRV (A) and RMSSD (B) relative to baseline (*ΔHRV) in meditation and phone conditions across three time points.*

interaction, $F(1.95, 328.91) = .490$, $p = .61$. partial $\eta^2 = .003$ (see Table 2). Consistent with HF-HRV analysis, results indicate that participants assigned to the meditation condition exhibited relatively greater increases in RMSSD during the experimental period than the participants assigned to the phone condition with an effect size in the small-to-medium range (see Figure 1B). Unlike the HF-HRV analyses, the effect of time was not statistically significant.

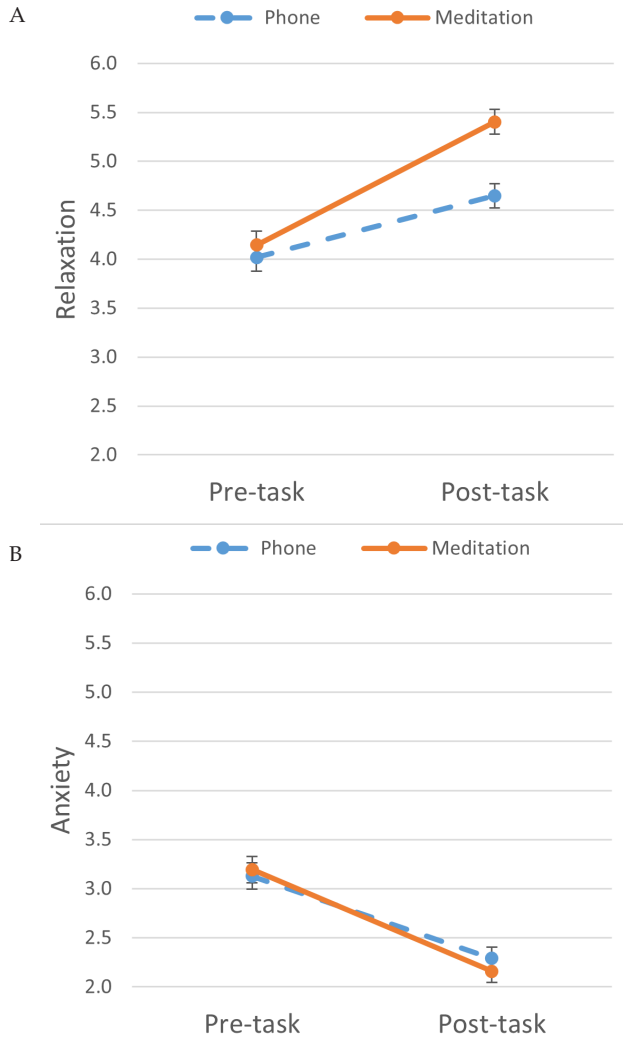
SUBJECTIVE MOOD

In examining relaxation as a dependent variable, results indicated a significant main effect of time, $F(1, 169) = 149.49$, $p < .001$, partial $\eta^2 = .47$, and condition, $F(1, 169) = 6.70$, $p = .01$, partial $\eta^2 = .038$, as well as Condition × Time interaction, $F(1, 169) = 16.58$, $p < .001$, partial $\eta^2 = .089$ (see Table 2). These results indicate that relaxation increased substantially in both conditions; however, participants assigned to the meditation condition exhibited relatively greater increases in relaxation following the experimental period than the participants assigned to the phone condition (see Figure 2A). The effect of time was large, the main effect of condition was small, and Condition × Time effect was medium.

In an exploratory analysis examining anxiety as a dependent variable, results indicated a significant main effect of time, $F(1, 169) = 177.93$, $p < .001$, partial $\eta^2 = .51$, but neither condition, $F(1, 169) = .04$, $p = .84$, partial $\eta^2 < .001$, nor Condition × Time interaction, $F(1, 169) = 1.99$, $p = .16$, partial $\eta^2 = .012$ (see Table 2). These results indicate that anxiety decreased substantially in both conditions; however, the degree of change was similar across both conditions (see Figure 2B). The effect of time was large and the effect of Condition × Time effect was small.

DISCUSSION

Mindfulness practice has been proposed as a promising alternative to excessive engagement with mobile phones (Brewer, 2019; Kabat-Zinn, 2005). Prior correlational studies have highlighted the association of dispositional mindfulness with negative mental health impacts of phone and social media use (Feldman et al.,



Note. Graphed values refect estimated marginal means and bars represent one standard error.

FIGURE 2. Change in subjective relaxation (A) and anxiety (B) in meditation and phone conditions from pre- to post-task.

2011; Jones et al., 2022; Liu et al., 2017); however, the immediate effects of mindful states evoked by meditation practice and typical use of mobile phones have not been directly compared in laboratory research to date.

The present experiment randomly assigned 171 university students to one of two 15-minute activities. In the unstructured screen-time condition, participants were prompted to use their phones in a manner consistent with their typical daily use. In the meditation condition, participants were prompted to follow a recorded guided meditation focused on mindfulness of the breath. Consistent with hypotheses, individuals in the meditation condition demonstrated greater increases in heart rate variability (HRV). The increase in HRV in the meditation condition can be interpreted as evidence of a relaxation response (Nakao, 2019) but also as a potential biomarker of emotion- and attention-regulation process (Appelhans & Luecken, 2006), a proposed mechanism of the salutary effects of mindfulness practice (Christodoulou et al., 2020). To provide some context, the effect sizes of the main effects of condition on HRV (small-to-medium for RMSSD, medium-to-large for HF-HRV) are within the range of meta-analytically derived effect sizes of associations of HRV and other positive psychological factors including compassion (Di Bello et al., 2020) and self-control (Zahn et al., 2016). In terms of subjective measures of emotional state, the mindfulness condition was found to be superior to phone use in terms of increasing states of relaxation, paralleling the results of the HRV analyses. An exploratory analysis revealed that the two activities were equally effective at reducing state anxiety. These findings suggest that meditation has an advantage over mobile phone use for promoting states of relaxation although both activities may offer short-term reductions in unpleasant emotional states.

IMPLICATIONS

These results have implications for promoting behavior change among individuals wishing to replace excessive mobile phone use with alternative habits such as mindfulness practice. The finding that mindfulness meditation practice improved HRV and relaxation more than equivalent time spent on typical mobile phone screen time activities could be highlighted in informational

materials for people considering reducing their phone use. As an analogy, educational materials for smokers emphasize both immediate cardiovascular benefits of quitting smoking (e.g., decreased heart rate within minutes) as well as benefits that are observable over weeks, months, and years (e.g., Centers for Disease Control and Prevention, 2020). Although the present study compared 15 minutes of meditation and phone use, the differential benefits of meditation over phone use for HRV were evident after only 5 minutes, suggesting that even a brief switch in activities may have significant short-terms effects. The present findings could be coupled with results from studies of longer duration (Hunt et al., 2018; Tromholt, 2016) in educational materials to demonstrate short- and long-term mental health benefits that may result from reducing digital activities such as social media use.

In addition to these potential applications of this finding, this result also adds to the literature on the relationship between HRV and the construct of mindfulness by replicating and extending the finding that laboratory induction of mindful states produces greater increases in HRV than various control conditions including authentic (Ditto et al., 2006, Study 1) and sham (May et al., 2016) relaxation procedures as well as exposure to conceptual presentations of mindfulness without practice (Azam et al., 2015). Ditto and colleagues (2006, Study 2) found mindfulness practice resulted in greater HRV increases than listening to excerpts from a popular audio book. The present study conceptually replicates and extends this prior finding by examining a more widely practiced leisure activity (i.e., unstructured screen time in a manner selected by participants), adding an increased level of ecological validity to this contrast. Taken together, these laboratory studies also complement randomized-controlled studies of mindfulness cultivated via longer mindfulness-based interventions (Brown et al., 2021; Christodoulou et al., 2020) by isolating the HRV effects of mindfulness practice specifically from the variety of other potentially therapeutic factors typically present in multi-component mindfulness training programs (Feldman et al., 2010, Keng et al., 2011).

The present study has several methodological strengths including an experimental design integrating objective and subjective assessments of relaxation, the use of a previously validated mindfulness induction procedure (Feldman et al., 2010), and adherence to guidelines for best practices in the design, data

collection, and reporting of HRV research (Laborde et al., 2017). Nonetheless, there are also limitations in this study that deserve mention in terms of their implications for both interpreting the results and informing future research.

First, the use of an entirely female-identifying sample of university students introduces both strengths and limitations. On the one hand, the relevance of these findings to this specific population are noteworthy in that the benefits of mindfulness-based interventions have been demonstrated to be of value for university students specifically (Dawson et al., 2020). The potential costs of excessive phone use for young women specifically has been suggested by the relatively stronger association of screen time and poor mental health among young women (Twenge et al., 2018). On the other hand, the generalizability of these results to different populations is limited, particularly in light of documented differences in HRV associated with gender (Koeing & Thayer, 2016) and ethnicity (Hill et al., 2015). Because the present sample was primarily White, non-Hispanic, and entirely female identifying, it would be important to test the replicability of these results in different samples with greater diversity in age and gender, as well as ethnic and racial identities.

A second limitation was that we did not assess potential mechanisms that may explain differences in HRV change across mindfulness meditation and mobile phone screen time. One potential candidate would be patterns of respiration, which can influence HRV values via respiratory sinus arrythmia (Laborde et al., 2017). Instructions to regulate breathing patterns can result in increased HRV and are commonly employed in stress management techniques (McCarty & Shaffer, 2015). The meditation instructions used in the present study did not ask participants to adjust their breathing but to simply observe the sensations associated with breathing. Nonetheless, it is possible that simply drawing attention to the sensations of breathing may have influenced respiratory activity in ways not present during the phone condition. Furthermore, prior research on mobile phone use in laboratory contexts suggests that the typical posture assumed when using a smartphone can have an acute effect on respiratory function (Kang et al., 2016). Measuring respiration was beyond the scope of the present study. To address this issue, we included RMSSD as an alternative metric of HRV that is less influenced by respiration than HF-HRV (Laborde et al., 2017). The replication of the findings

across both metrics offers some indication that the hypothesized effects are relatively robust to the effects of respiration; however, the relatively smaller effect size of the RMSSD findings suggests that respiration may indeed explain a portion of this observed effect. A final conceptual mechanism that may explain the differential effects of the two activities on HRV and relaxation would be emotion regulation (Appelhans & Luecken, 2006; Williams et al., 2015). As such, self-report and task-based measures of this construct would be fruitful to examine in future research.

Finally, it is important to note that the results of this study speak only to the immediate effects of meditation and phone use. Although there is a growing body of studies examining HRV following participation in longer mindfulness-based interventions (Brown et al., 2021; Christodoulou et al, 2020), there is not an analogous literature that has examined the longitudinal effects of experimentally manipulated mobile phone use on cardiovascular outcomes. We hope that the present findings will provide impetus for studies comparing more substantial and sustained interventions to support the practice of mindfulness meditation and the reduction of excessive or problematic mobile phone use.

CONCLUSIONS

In closing, it is important to note that while many mobile phone users may be looking to cut down the time they spend on these devices (Deloitte, 2018; Jiang, 2018), relatively fewer are looking to replace this activity entirely, nor do many users necessarily find all aspects of mobile phone use objectionable or personally harmful. The present study frames mindfulness meditation as an alternative activity to unstructured mobile phone screen time and offers evidence that the former may be more physiologically and emotionally restorative than the latter. A promising area for future inquiry may be how best to combine these two activities in daily life to examine whether periodically taking a break from screen time for brief mindfulness practices can enhance the psychological experience of subsequent smartphone use. Applying proposed mechanisms of mindfulness-based interventions in addictive behaviors (Brewer, 2019), it is possible that taking a

break to observe present-moment experiences with openness and curiosity may help heighten awareness as to which subsequent phone activities are personally nourishing and which are depleting. This may in turn facilitate more intentional choices of how one spends time on their phone. Similarly, a pause may also invite users to reallocate attention away from phones when there are opportunities for rewarding experiences in the user's immediate, non-digital environments such as in-person interactions, spending time outdoors, or savoring a meal. A pause from the distracting stimulation of screens to observe inner experiences may also facilitate emotion regulation by allowing the user to become aware of thoughts, feelings, and sensations of fatigue, loneliness, and anxiety that can motivate more effective and targeted coping responses (Kabat-Zinn, 2005). The nonjudgemental awareness inherent in mindfulness practice may also help users navigate social media with less self-criticism (Jones et al., 2022) or reactivity to harsh judgments expressed online by others. In sum, future research can help address whether mindfulness offers a more sustainable way of being in the digital world.

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